

Chapter 3

Water and the Fitness of the Environment

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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Overview: The Molecule That Supports All of Life

- Water is the biological medium on Earth
- All living organisms require water more than any other substance
- Most cells are surrounded by water, and cells themselves are about 70–95% water
- The abundance of water is the main reason the Earth is habitable

Fig. 3-1

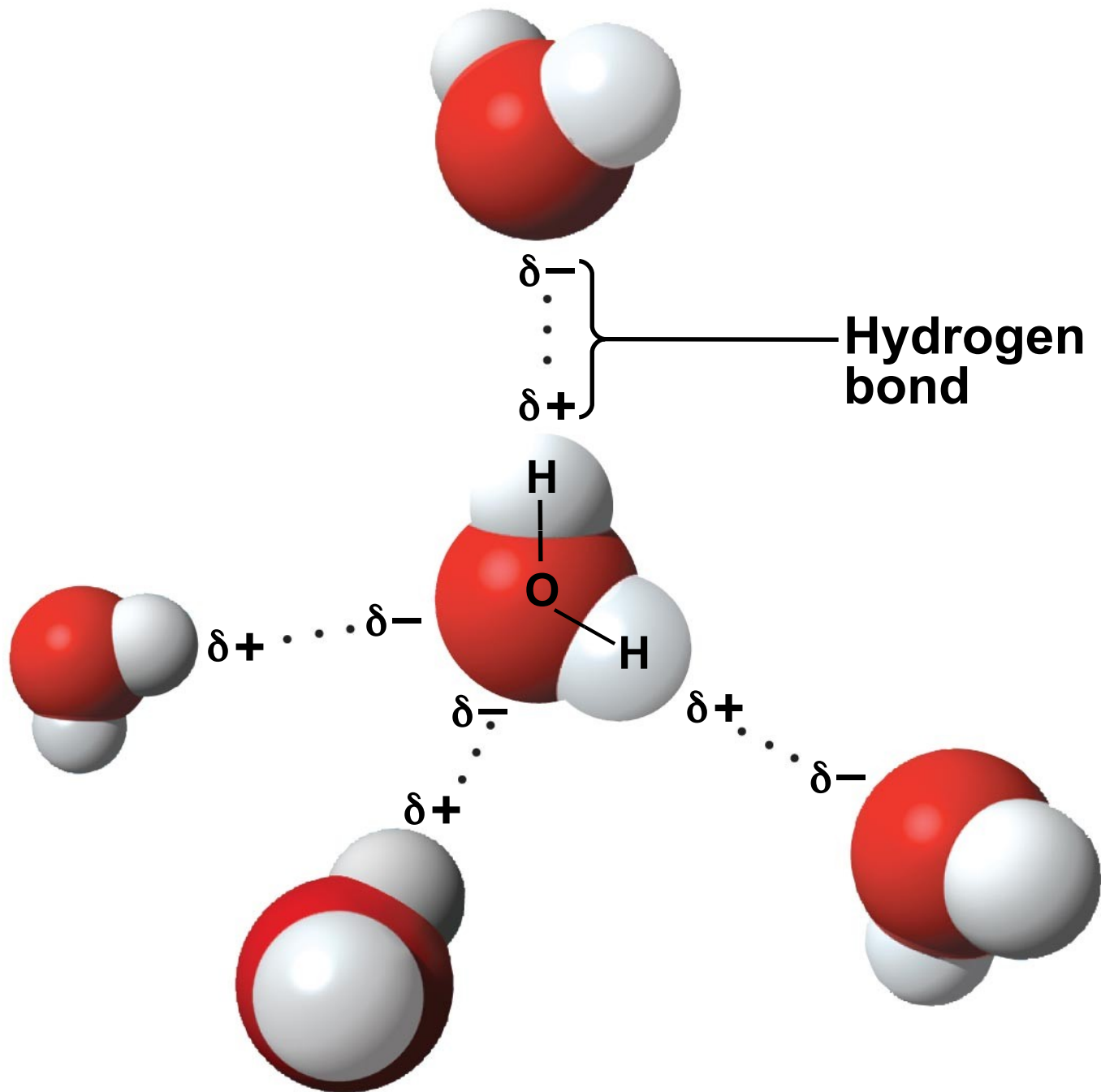


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Concept 3.1: The polarity of water molecules results in hydrogen bonding

- The water molecule is a **polar molecule**: The opposite ends have opposite charges
- Polarity allows water molecules to form hydrogen bonds with each other

Fig. 3-2



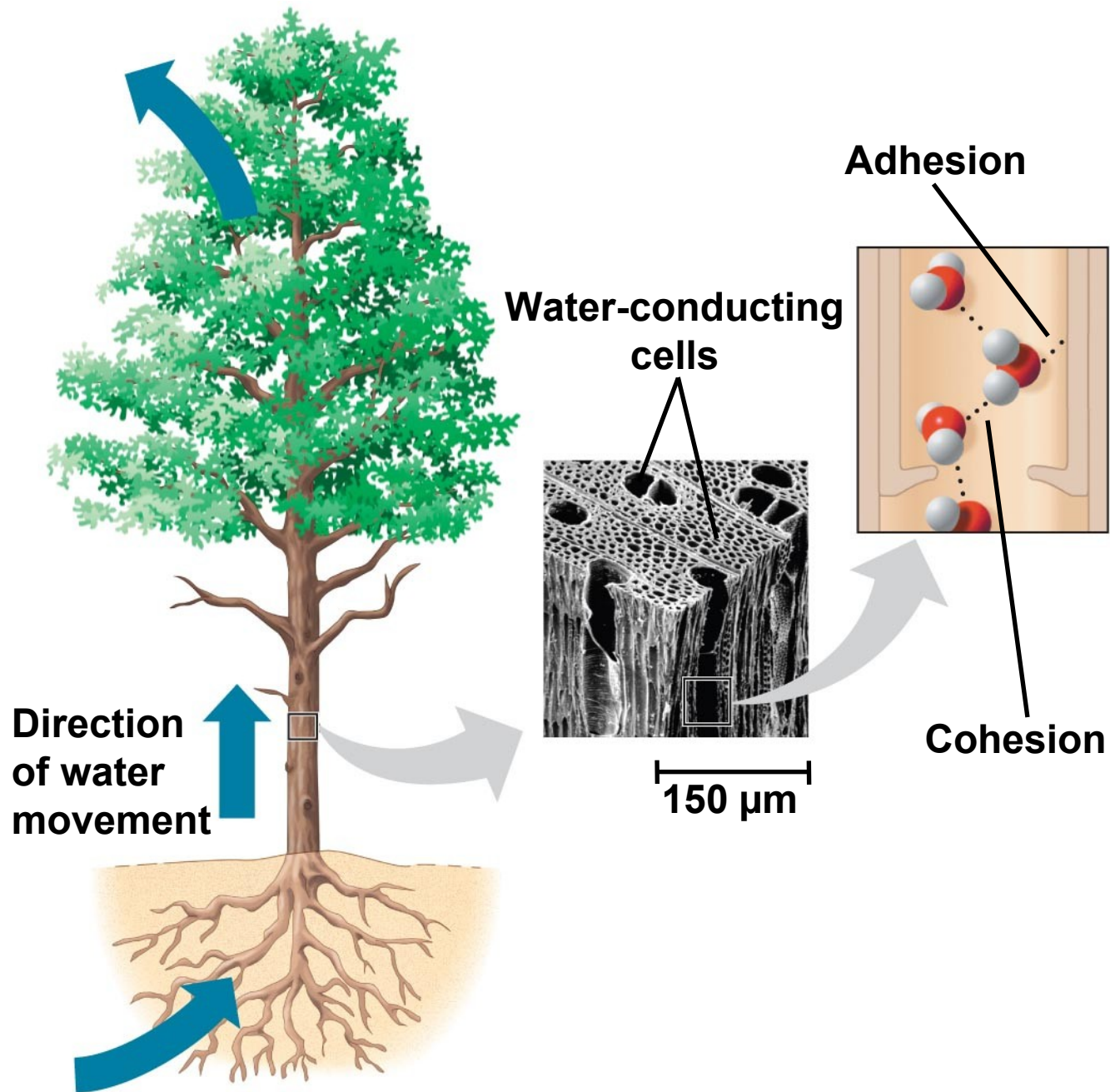
Concept 3.2: Four emergent properties of water contribute to Earth's fitness for life

- Four of water's properties that facilitate an environment for life are:
 - Cohesive behavior
 - Ability to moderate temperature
 - Expansion upon freezing
 - Versatility as a solvent

Cohesion

- Collectively, hydrogen bonds hold water molecules together, a phenomenon called **cohesion**
- Cohesion helps the transport of water against gravity in plants
- **Adhesion** is an attraction between different substances, for example, between water and plant cell walls

Fig. 3-3



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- **Surface tension** is a measure of how hard it is to break the surface of a liquid
 - Surface tension is related to cohesion

Fig. 3-4



Moderation of Temperature

- Water absorbs heat from warmer air and releases stored heat to cooler air
- Water can absorb or release a large amount of heat with only a slight change in its own temperature

Heat and Temperature

- **Kinetic energy** is the energy of motion
- **Heat** is a measure of the *total* amount of kinetic energy due to molecular motion
- **Temperature** measures the intensity of heat due to the *average* kinetic energy of molecules

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- The **Celsius scale** is a measure of temperature using Celsius degrees ($^{\circ}\text{C}$)
 - A **calorie (cal)** is the amount of heat required to raise the temperature of 1 g of water by 1°C
 - The “calories” on food packages are actually **kilocalories (kcal)**, where $1 \text{ kcal} = 1,000 \text{ cal}$
 - The **joule (J)** is another unit of energy where $1 \text{ J} = 0.239 \text{ cal}$, or $1 \text{ cal} = 4.184 \text{ J}$

Water's High Specific Heat

- The **specific heat** of a substance is the amount of heat that must be absorbed or lost for 1 g of that substance to change its temperature by 1°C
- The specific heat of water is $1 \text{ cal/g/}^{\circ}\text{C}$
- Water resists changing its temperature because of its high specific heat

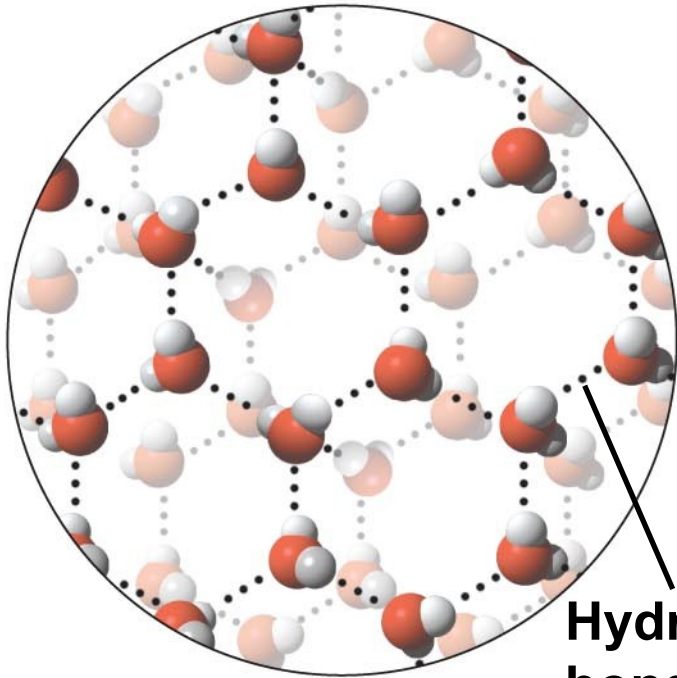
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- Water's high specific heat can be traced to hydrogen bonding
 - Heat is absorbed when hydrogen bonds break
 - Heat is released when hydrogen bonds form
 - The high specific heat of water minimizes temperature fluctuations to within limits that permit life

Evaporative Cooling

- *Evaporation* is transformation of a substance from liquid to gas
- **Heat of vaporization** is the heat a liquid must absorb for 1 g to be converted to gas
- As a liquid evaporates, its remaining surface cools, a process called **evaporative cooling**
- Evaporative cooling of water helps stabilize temperatures in organisms and bodies of water

Insulation of Bodies of Water by Floating Ice

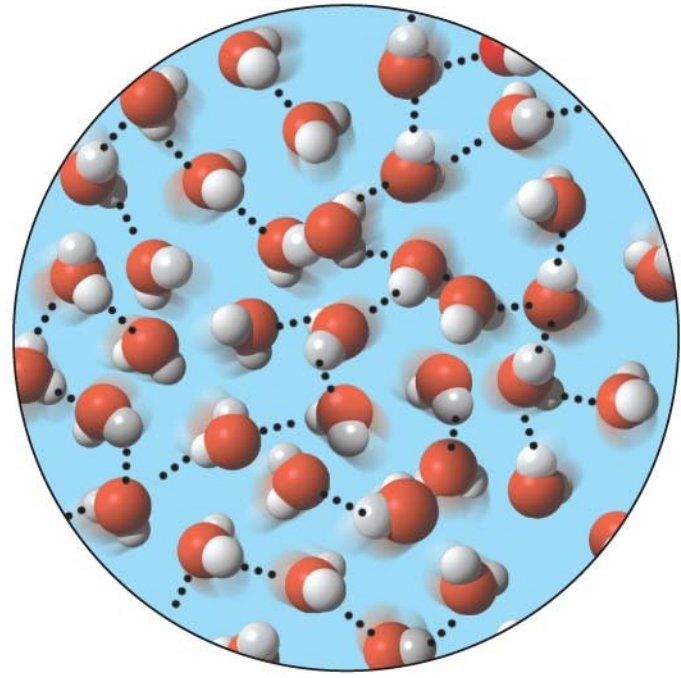
- Ice floats in liquid water because hydrogen bonds in ice are more “ordered,” making ice less dense
- Water reaches its greatest density at 4°C
- If ice sank, all bodies of water would eventually freeze solid, making life impossible on Earth



**Hydrogen
bond**

Ice

Hydrogen bonds are stable



Liquid water

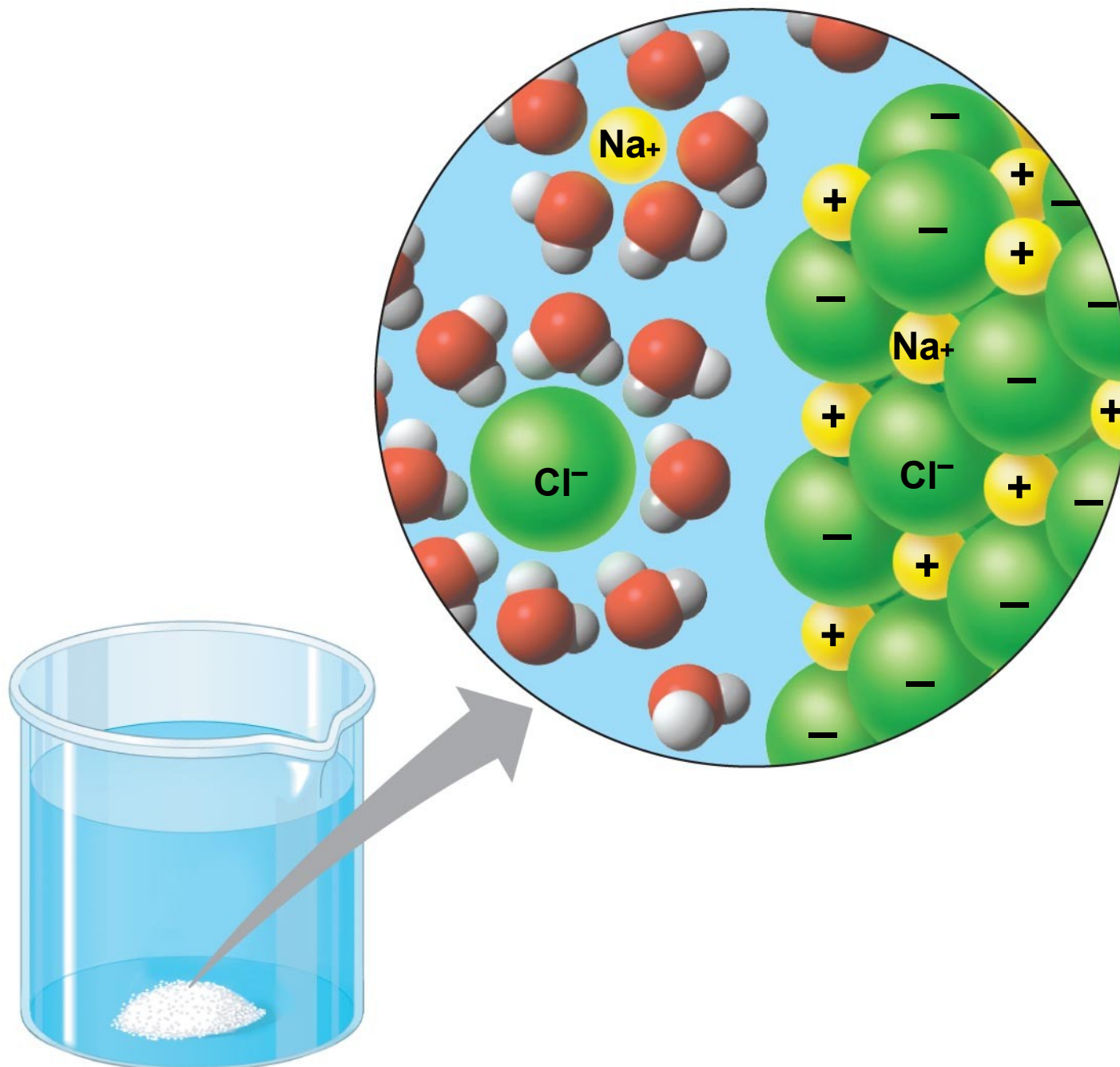
Hydrogen bonds break and re-form

The Solvent of Life

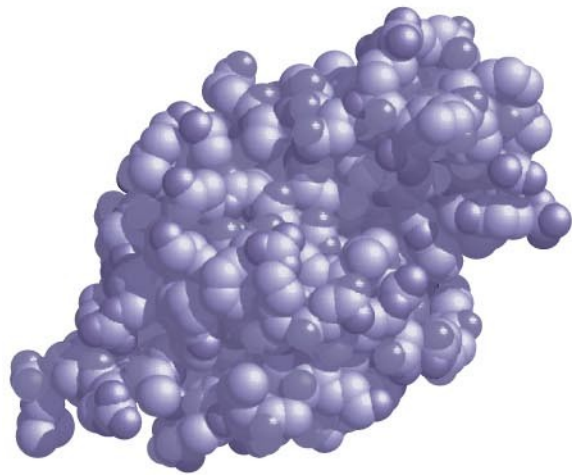
- A **solution** is a liquid that is a homogeneous mixture of substances
- A **solvent** is the dissolving agent of a solution
- The **solute** is the substance that is dissolved
- An **aqueous solution** is one in which water is the solvent

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- Water is a versatile solvent due to its polarity, which allows it to form hydrogen bonds easily
 - When an ionic compound is dissolved in water, each ion is surrounded by a sphere of water molecules called a **hydration shell**

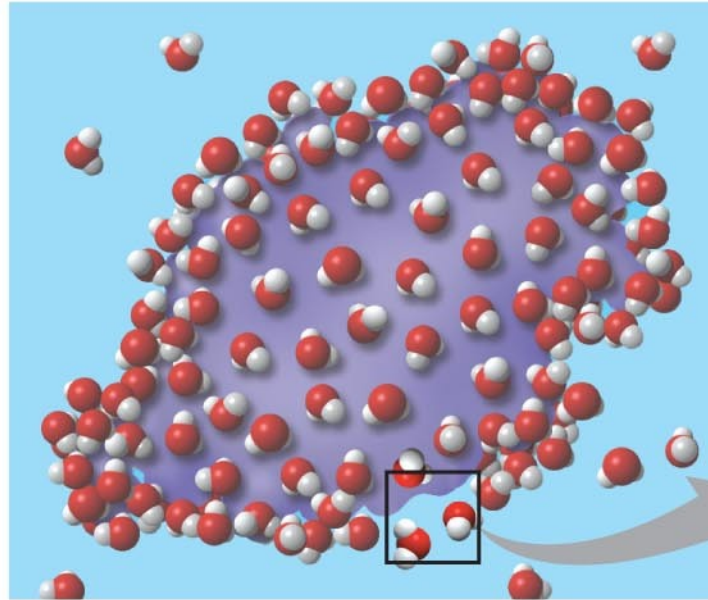
Fig. 3-7



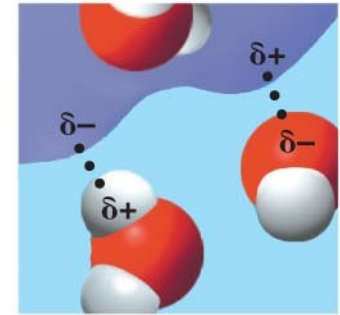
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- Water can also dissolve compounds made of nonionic polar molecules
 - Even large polar molecules such as proteins can dissolve in water if they have ionic and polar regions



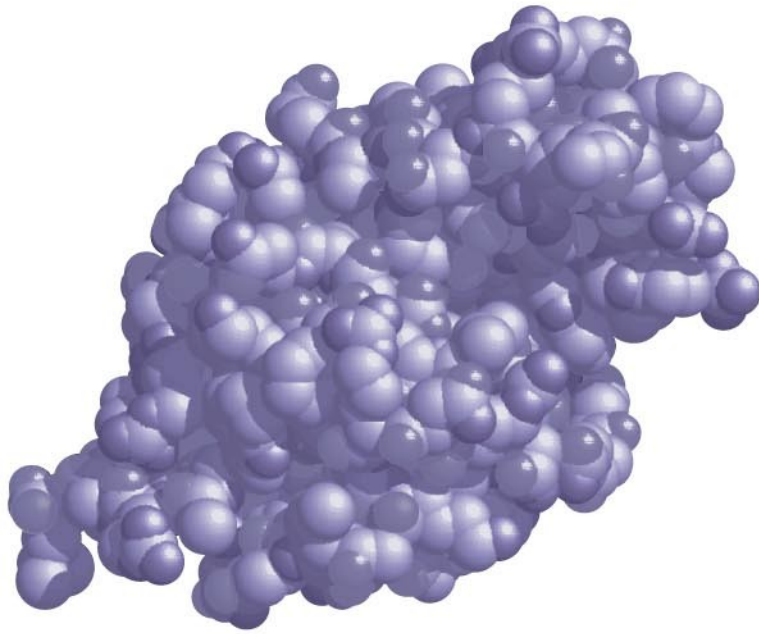
(a) Lysozyme molecule in a nonaqueous environment



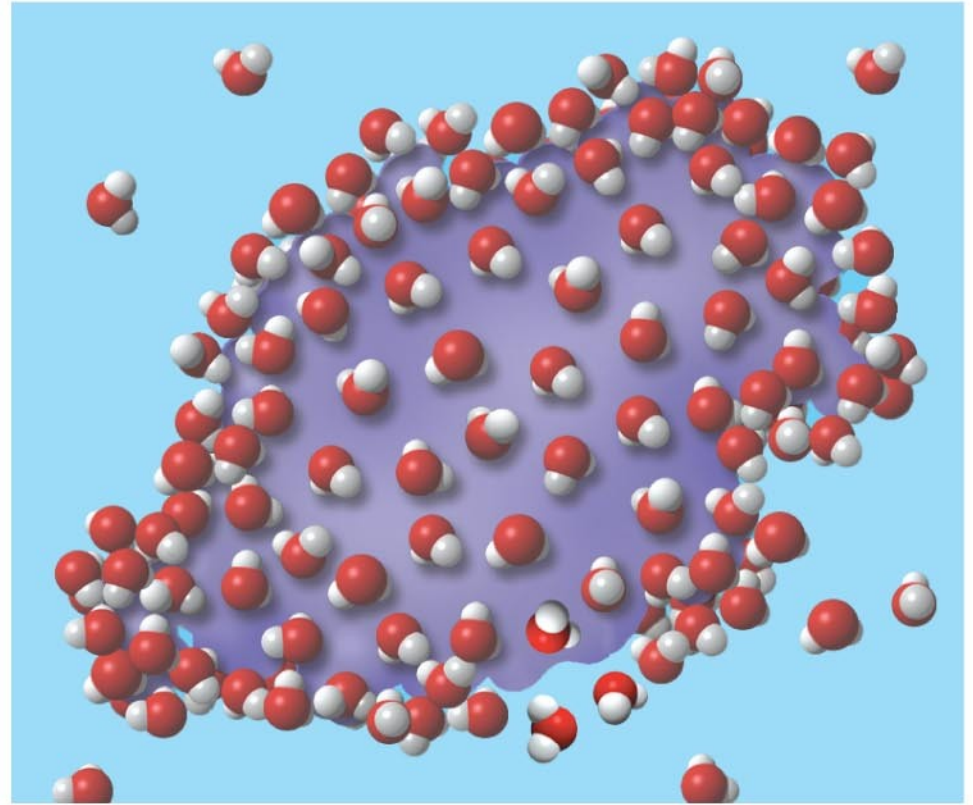
(b) Lysozyme molecule (purple) in an aqueous environment



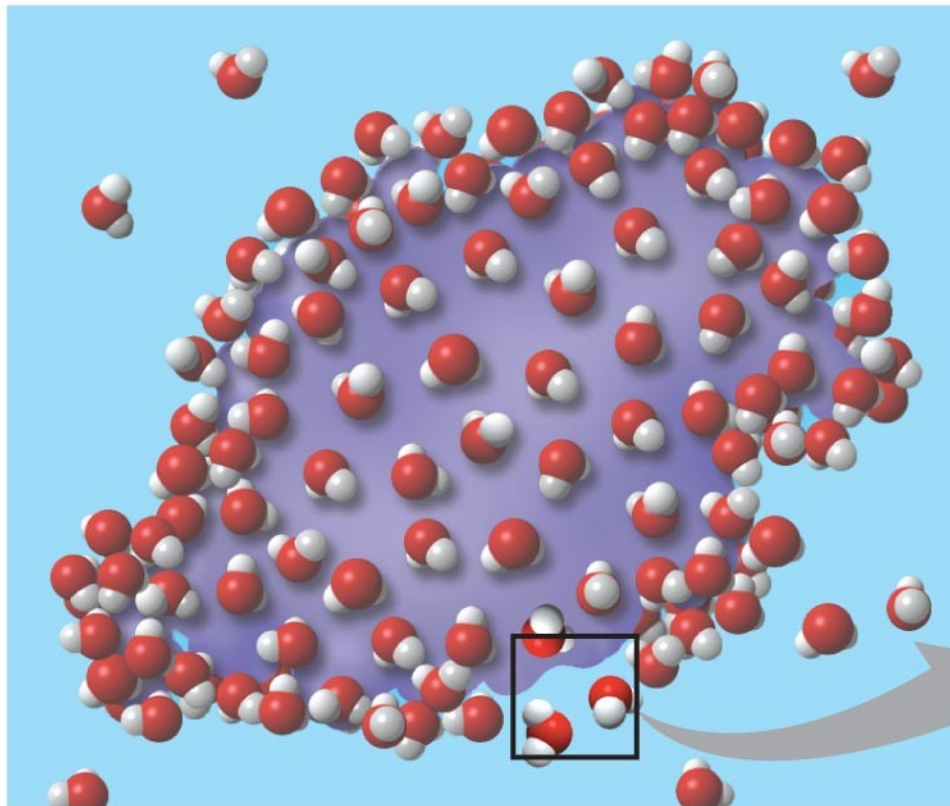
(c) Ionic and polar regions on the protein's surface attract water molecules.



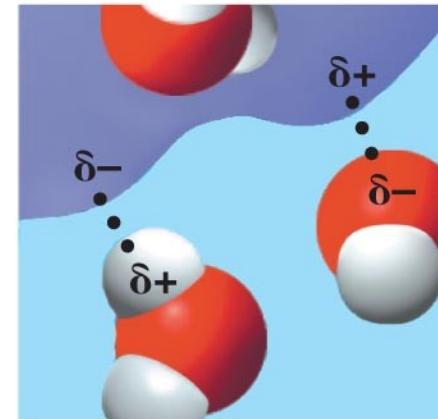
(a) Lysozyme molecule in a nonaqueous environment



(b) Lysozyme molecule (purple) in an aqueous environment



(b) Lysozyme molecule (purple) in an aqueous environment



(c) Ionic and polar regions on the protein's surface attract water molecules.

Hydrophilic and Hydrophobic Substances

- A **hydrophilic** substance is one that has an affinity for water
- A **hydrophobic** substance is one that does not have an affinity for water
- Oil molecules are hydrophobic because they have relatively nonpolar bonds
- A **colloid** is a stable suspension of fine particles in a liquid

Solute Concentration in Aqueous Solutions

- Most biochemical reactions occur in water
- Chemical reactions depend on collisions of molecules and therefore on the concentration of solutes in an aqueous solution

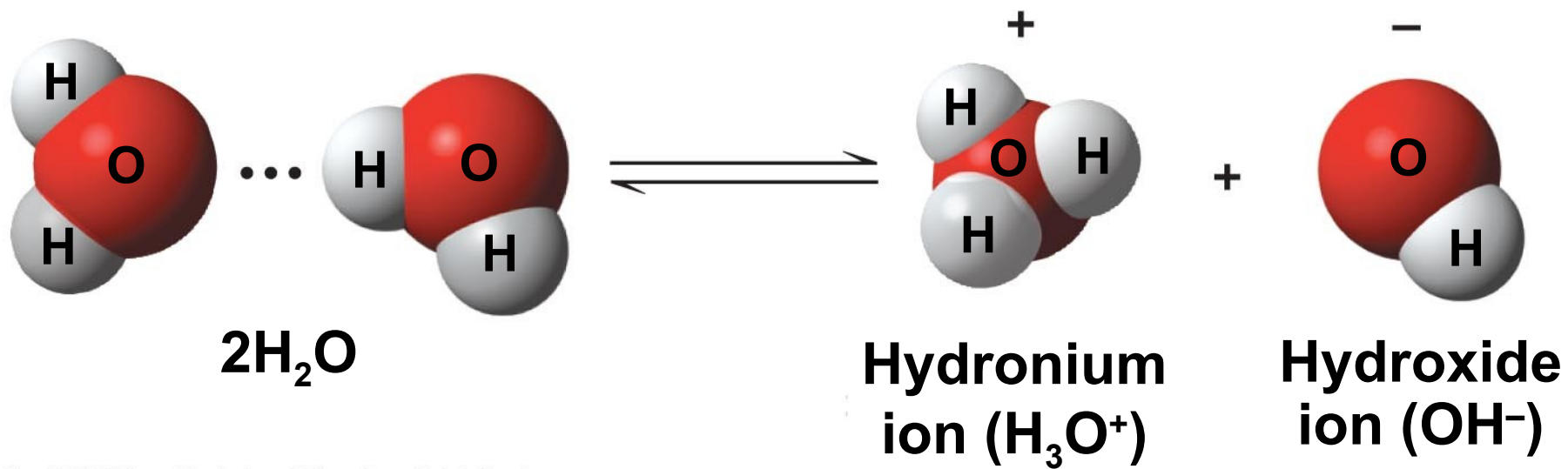
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- **Molecular mass** is the sum of all masses of all atoms in a molecule
 - Numbers of molecules are usually measured in moles, where 1 **mole (mol)** = 6.02×10^{23} molecules
 - Avogadro's number and the unit *dalton* were defined such that 6.02×10^{23} daltons = 1 g
 - **Molarity (*M*)** is the number of moles of solute per liter of solution

Concept 3.3: Acidic and basic conditions affect living organisms

- A hydrogen atom in a hydrogen bond between two water molecules can shift from one to the other:
 - The hydrogen atom leaves its electron behind and is transferred as a proton, or **hydrogen ion** (H^+)
 - The molecule with the extra proton is now a **hydronium ion** (H_3O^+), though it is often represented as H^+
 - The molecule that lost the proton is now a **hydroxide ion** (OH^-)

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- Water is in a state of dynamic equilibrium in which water molecules dissociate at the same rate at which they are being reformed

Fig. 3-UN2



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- Though statistically rare, the dissociation of water molecules has a great effect on organisms
 - Changes in concentrations of H^+ and OH^- can drastically affect the chemistry of a cell

Effects of Changes in pH

- Concentrations of H^+ and OH^- are equal in pure water
- Adding certain solutes, called acids and bases, modifies the concentrations of H^+ and OH^-
- Biologists use something called the pH scale to describe whether a solution is acidic or basic (the opposite of acidic)

Acids and Bases

- An **acid** is any substance that increases the H^+ concentration of a solution
- A **base** is any substance that reduces the H^+ concentration of a solution

The pH Scale

- In any aqueous solution at 25°C the product of H^+ and OH^- is constant and can be written as
 $[H^+]$

$$[OH^-] = 10^{-14}$$

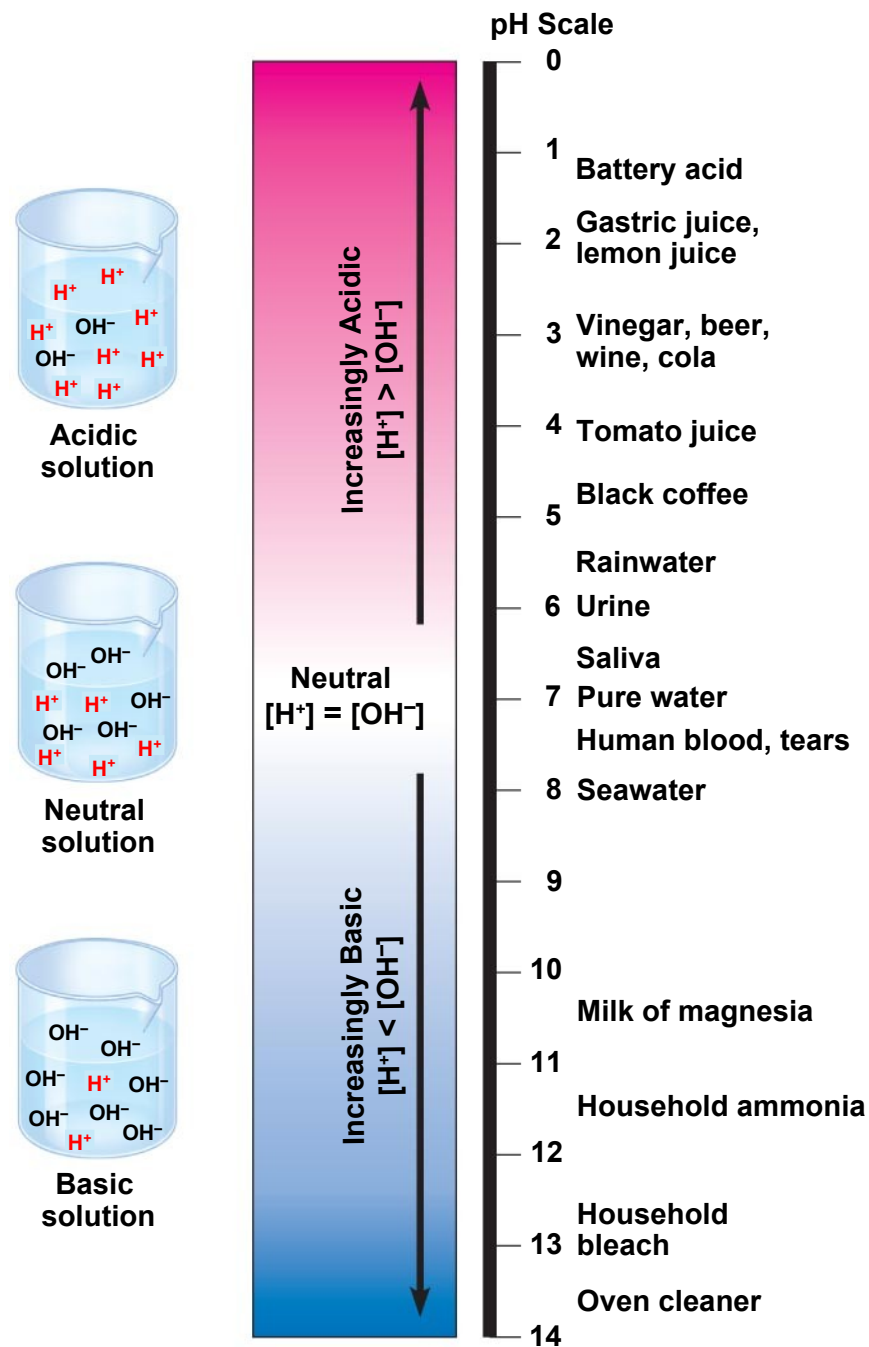
- The **pH** of a solution is defined by the negative logarithm of H^+ concentration, written as
$$pH = -\log [H^+]$$

- For a neutral aqueous solution

$$[H^+] \text{ is } 10^{-7} = -(-7) = 7$$

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- Acidic solutions have pH values less than 7
 - Basic solutions have pH values greater than 7
 - Most biological fluids have pH values in the range of 6 to 8

Fig. 3-9



Buffers

- The internal pH of most living cells must remain close to pH 7
- **Buffers** are substances that minimize changes in concentrations of H^+ and OH^- in a solution
- Most buffers consist of an acid-base pair that reversibly combines with H^+

Threats to Water Quality on Earth

- **Acid precipitation** refers to rain, snow, or fog with a pH lower than 5.6
- Acid precipitation is caused mainly by the mixing of different pollutants with water in the air and can fall at some distance from the source of pollutants
- Acid precipitation can damage life in lakes and streams
- Effects of acid precipitation on soil chemistry are contributing to the decline of some forests

Fig. 3-10



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- Human activities such as burning fossil fuels threaten water quality
 - CO₂ is released by fossil fuel combustion and contributes to:
 - A warming of earth called the “greenhouse” effect
 - Acidification of the oceans; this leads to a decrease in the ability of corals to form calcified reefs

You should now be able to:

1. List and explain the four properties of water that emerge as a result of its ability to form hydrogen bonds
2. Distinguish between the following sets of terms: hydrophobic and hydrophilic substances; a solute, a solvent, and a solution
3. Define acid, base, and pH
4. Explain how buffers work